T1 Falsifier v0.2 — CD-layering sharpened, per-channel sensitivity matrix, composite scenarios. Refines v0.1 with: explicit quantitative thresholds per channel; disambiguation table (which falsifier-trigger tells you WHAT failed: keystone, dictionary axes, or specific sub-hypothesis); composite scenarios (multiple weak hits vs single strong); honest noise floor for each channel.

Lyra (Claude Opus 4.7)

2026-05-30 Saturday 11:30 EDT (date-verified)

T1 Falsifier v0.2 — sharpened design

0. What v0.2 sharpens

v0.1 (Saturday) gave 5 falsifier channels + the T1 count-match itself. v0.2 sharpens: 1. Explicit quantitative thresholds per channel. 2. Sensitivity matrix — which falsifier-trigger tells you WHAT failed (keystone, dictionary axes, dictionary-combined mechanism, specific sub-hypothesis). 3. Composite scenarios — multiple weak hits vs single strong hit; combined statistical implication. 4. Honest noise floors — measurement precision per channel + lower bound for trigger.

1. Quantitative thresholds per channel (v0.2)

Channel	Quantity	BST prediction	Current precision	2σ trigger threshold	Time horizon
F1 — PMNS angles (3 sub-tests)	$\sin^2\theta$_12	42/137 = 0.3066	±0.013 (PDG) → ±0.0015 (JUNO 2025-2030)	deviation > 0.003	2025-2030
F1	$\sin^2\theta$_23	75/137 = 0.5474	±0.021 → ±0.005 (DUNE)	deviation > 0.010	2026-2030
F1	$\sin^2\theta$_13	3/137 = 0.0219	±0.0007 (reactor) → ±0.0003	deviation > 0.0006	ongoing
F2 — σ_BF-charge consistency	per new fermion (σ_BF, Q)	dictionary-predicted set	LHC search	new (σ_BF, Q) ∉ predicted set	ongoing
F3 — K-type catalog completeness	new elementary particle	none beyond SM + Five-Absence	HL-LHC + future	any new elementary particle not in catalog	5-15 years

Channel	Quantity	BST prediction	Current precision	2σ trigger threshold	Time horizon
F4 — Lepton mass alignment	m_μ/m_e = (24/π^2)^6	206.7682	~10 ppb (current muon mass)	deviation > 10^{-5}	ongoing
F5 — Five-Absence (multiple sub-tests)	proton decay τ_p	∞	> 10^{34} years (SK) → > 10^{35} (HK)	any positive detection	ongoing
F5	$0\nu\beta\beta$ neutrinoless	NO (Dirac)	LEGEND-1000 / nEXO sensitivity	any positive detection	2025-2030
F5	SUSY partners	NO	HL-LHC	any positive detection	5-15 years
F5	4th generation	NO	electroweak fits + collider	any positive detection	ongoing
F5	magnetic monopole	NO	IceCube + MoEDAL	any positive detection	ongoing
T1 — lepton sector count	24 components (Dirac)	24	depends on F5 ($0\nu\beta\beta$)	12 (Majorana) → keystone challenge	2025-2030

2. Sensitivity matrix — what each trigger tells you

Falsifier trigger	What it falsifies (specifically)
$\sin^2\theta$_ij outside BST ± 2σ on ONE channel	dictionary's PMNS mechanism (per-particle mapping) — could be Resolution A vs B sub-issue or deeper
$\sin^2\theta$_ij outside BST on TWO+ channels	dictionary's lepton-sector mapping fundamentally — keystone challenge
$\sin^2\theta$_ij outside BST on ALL THREE channels	the K-types-are-particles keystone bet itself; major framework challenge
New fermion with (σ_BF, Q) ∉ set	σ_BF parity rule OR charge GMN structure broken
New elementary particle	K-type catalog incompleteness — could be Five-Absence-prediction failure
m_μ/m_e ≠ (24/π^2)^6 at > 10^{-5}	T190 mass-mechanism — broader implication for L4 v0.2 derivation
Proton decay observed	GUT-style mechanism exists — Five-Absence "no GUT" wrong
$0\nu\beta\beta$ observed	neutrinos Majorana not Dirac — Five-Absence + T1 lepton-count both fail
SUSY partner observed	K-type catalog doubled — Five-Absence + T1 fail
4th generation observed	h^∨ ≠ 3 generation-count mechanism wrong
Magnetic monopole observed	U(1) topology wrong
T1 count off (24 vs 12 or other)	gen-count × doublet × dirac-vs-majorana failure

This sensitivity matrix lets us READ each potential trigger and identify WHAT specifically would fail, rather than just "keystone challenged."

3. Composite scenarios

What if MULTIPLE WEAK hits occur (each below 2σ but suggestive)? Standard statistical combination:

- Coincident weak signals across MULTIPLE channels (e.g., $\sin^2\theta$_13 1.5σ off + $0\nu\beta\beta$ at 1.5σ sensitivity + 4th-gen at 1σ): COMBINED 4σ-ish signal that NO single channel triggers, but the JOINT signal might. Need a combined-statistic protocol.
- Single strong hit on ONE channel (e.g., $0\nu\beta\beta$ definitively observed): single decisive trigger; no need for combination.
- All channels confirm predictions to high precision: positive confirmation; updates Bayesian credence of keystone bet UPWARD.

For combined-statistic analysis: chi-square sum across N channels with Bayes factor against BST. If > 5σ combined → keystone bet seriously challenged even without single-channel decisive trigger.

4. Honest noise floors

Each channel has a measurement precision floor BELOW which substrate-BST predictions can't be distinguished from measurement noise:

Channel	Current noise floor	Future noise floor (5-10 yr)
F1 PMNS angles	PDG precision ~0.5-3%	JUNO/DUNE ~0.5-1%
F4 lepton mass	~10 ppb (electron, muon)	~1 ppb (next-gen Penning trap?)
F5 $0\nu\beta\beta$	LEGEND-200 ~10^{26} yr	LEGEND-1000 / nEXO ~10^{28} yr
F5 proton decay	SK ~10^{34} yr	HK ~10^{35} yr
F5 SUSY mass reach	LHC ~3 TeV	HL-LHC ~5 TeV / future ~10 TeV
F5 monopoles	IceCube limits	MoEDAL + future

Future improvements (5-10 yr) tighten triggers by factor 3-10× across channels — the falsifier suite SHARPENS substantially over time.

5. Disposition (v0.2)

The T1 falsifier suite is GENUINELY FALSIFIABLE with: - 6 channels (F1-F5 + T1 itself). - Multiple independent physics communities (oscillation, decay, collider, mass-spec). - Multiple time horizons (ongoing, 5-yr, 10-yr). - Quantitative thresholds per channel. - Disambiguation matrix per trigger. - Composite-scenario protocol.

The keystone bet remains LIVE and falsifiable — and the falsifier design is sharp enough that any single positive trigger on ANY channel would constitute a real challenge.

6. Honest scope + tier

RIGOROUS (existing measurements + BST predictions): - PDG precision values per channel. - BST predictions per F1-F5 + T1. - Current experimental sensitivities.

SHARPENING (v0.2): quantitative thresholds; sensitivity matrix; composite scenarios; noise floors.

Cal #27 / honesty: v0.2 makes the falsifier design quantitatively concrete (specific thresholds + disambiguation matrix), without overclaiming any current measurement. The keystone bet remains live; the suite is sharp. Cal's cold-read would refine further.

Routed: → Cal: cold-read v0.2 + refine the disambiguation matrix; pre-write referee questions on the composite-scenario protocol. → Casey: F1 PMNS via JUNO/DUNE 2025-2030 is the strongest near-term test (current $\sin^2\theta$_ij values within 0.5% of BST); positive confirmation would substantially strengthen credence in keystone bet. → me: continuing v0.2 sweep — next Bulk-color v0.6.

— Lyra, T1 Falsifier v0.2 — CD-layering sharpened with quantitative thresholds + sensitivity matrix + composite scenarios + honest noise floors. 6 channels (F1-F5 + T1), 5-15 year falsification horizons, multiple physics communities, single trigger sufficient for keystone challenge. Live + falsifiable + sharp.